Align and Complete Samples in Remote Sensing Fine-Grained Rigid Object Detection

Zicong Zhu, *Member, IEEE*, Jian Kang, *Member, IEEE*, Wenhui Diao, *Member, IEEE*, Bing Wang, *Member, IEEE*, and Jingen Ni, *Senior Member, IEEE*

***Abstract*—Currently, the remote sensing fine-grained rigid object detectors (RSFRODs) mainly face two challenges: fuzzy localization and inaccurate classification (including misclassifying and multiclassifying). First, mainstream detectors based on the "dense prediction" paradigm suffer from a misalignment between anchor points (APs) and ground truths (GTs). Commonly, they adopt multistage regression as a solution, which brings an ambiguous sample definition problem and redundant computational costs. Second, two factors seriously affect the classification. They are the insufficient sample learning caused by the long-tail distribution of categories, and the difficulty in extracting discriminative features caused by slight interclass variance. To address the issues above, we propose an efficient aligning and completing detector (ACDet) based on a single-stage structure. First, the adaptive anchor alignment mechanism (A^3M) decouples the centripetal sampling bias in the categorical features and leverages it to learn the APs aligned with GTs. It is plug-and-play, requiring no additional supervision annotations. Second, a novel online tail-sample supplementation (OTS) algorithm is proposed. It dynamically maintains class balance during training and can be easily added as postprocessing behind existing sample assignment strategies. Third, an adaptive group perceptron is designed to effectively enrich the diversity of features and enhance the model's ability to extract discriminative features. Experiments on four public datasets demonstrate that the proposed ACDet achieves the state-of-the-art (SOTA) level, even surpassing competitive multistage detectors, with fewer computing resources and a faster inference speed. The code will be public after the article is published.**

***Index Terms*—Feature diversification, fine-grained recognition, localization alignment, long-tail distribution, object detection, rigid object.**

I. Introduction

REMOTE sensing fine-grained rigid object detection (RSFROD) is an essential and fundamental visual interpretation task in Earth observation scenes [1], [2], [3], [4], [5]. Rigid objects refer to artificial targets on the ground, usually with stable and regular apparent structures [6], [7], [8], [9]. Detecting the above targets includes two subtasks: localization and classification. In localization tasks, oriented bounding boxes (OBBs) are commonly used to represent rigid targets accurately. The classification tasks aim to identify finer-grained subcategories based on various target usages. The RSFROD task has broad application values in traffic monitoring, fisheries management, Earth observation, and environmental protection [10], [11], [12], [13].

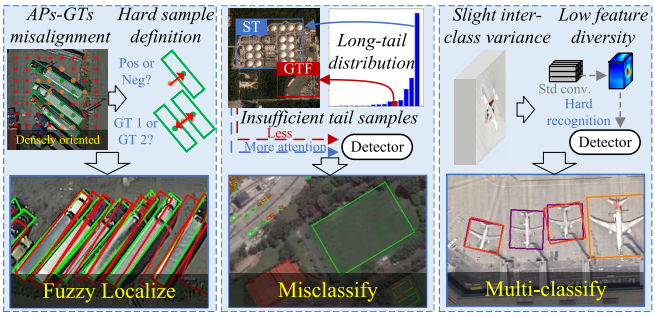


Fig. 1. Three major challenges in the RSFROD task. We systematically analyze the reasons for fuzzy localization, misclassification, and multiclassification. Ultimately, they are attributed to the misalignment between APs and GTs, incomplete samples, and low feature diversity during model training, respectively.

Currently, the RSFROD task faces three main challenges (see Fig. 1). First, the dense arrangement of arbitrarily oriented rigid objects [6], [7], [8] exacerbates the misalignment between anchor points and ground truths (APs-GTs). It leads to a hard sample definition and further results in fuzzy localization results. Second, the long-tail distribution of fine-grained objects causes incomplete sample assignment [2], [14]. This results in insufficient tail samples and, consequently, category recognition errors. Finally, the slight interclass variance of fine-grained objects reduces the diversity of features extracted by the model [15]. It affects the learning of discriminative features, which further leads to the coexistence of multiple classification results. The above issues seriously reduce the model's ability to localize and classify objects in the RSFROD task.

Regarding challenge 1, existing solutions can be divided into two streams. First, multistage regression based on dense prediction is an effective approach for RSFROD tasks [7], [16], [17], [18]. The core idea is to split the localization learning process into multiple stages. By proposing a multistage regression mechanism to refine the prediction box gradually, the localization has been effectively improved. The mainstream

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RSFROD methods are generally constructed based on the above paradigm [14], [15], [19], [20]. However, the problem of misalignment still exists, which further leads to ambiguous sample definitions. In addition, the repetitive cascade structure increases the computational overhead of the model, resulting in low inference efficiency. Second, the DETR-based methods [21], [22] and the keypoint-based methods [23] provide two alternative solutions. They entirely abandon the densely arranged anchor mechanism and instead utilize a new paradigm. The former guides the model to learn a fixed number of learnable object queries, while the latter utilizes the idea of keypoint extraction and clustering. Although both methods adopt new learning paradigms, they have not become mainstream due to their extremely long training time and mediocre performance on RSFROD tasks.

Existing works have made several improvements to address the challenges 2 and 3. For the former, strategies such as data augmentation and sample balance [14], [20] have emerged gradually, enhancing the model's ability to attend to samples from various categories. However, these solutions require replacing existing parts in the model and may interfere with related components when adapted to other models. For the latter, feature banks and graph network modeling methods [6], [14] have been introduced into the model's feature learning of fine-grained categories. Although the above strategies effectively improve the model's ability to discriminate fine-grained targets, they also incur significant computational cost. Efficient and straightforward solutions remain to be explored.

To solve the problem of APs-GTs misalignment, we propose an adaptive anchor alignment mechanism (A^3M). Specifically, it comprises an adaptive aligning network (AAN) and an adaptive anchor generator (AAG). It generates adaptive anchors that vary with the randomly distributed targets at the cost of increasing a few parameters. A^3M effectively alleviates the misalignment problem between uniformly preset anchors and random positions of targets in RS images. It reduces the difficulty of the model for localization regression, thereby improving performance. Furthermore, this mechanism is plug-and-play and can be applied to most existing dense prediction heads.

To address the problems of incomplete samples and low feature diversity, we propose a sample-feature completion strategy (SCS). Specifically, it consists of an online tail-sample supplementation (OTS) and an adaptive grouping perceptron (AGP). OTS is a postprocessing process of the existing sample assignment strategy, dynamically balancing the sample categories at the image level. AGP is specifically used to enhance the proposed AAN, primarily through the adaptive selection of the range of discriminative feature extraction in the spatial dimension. The above two submodules introduce an acceptable level of computational complexity to the model, significantly enhancing the completeness of samples and features. This thereby improves the model's recognition accuracy for fine-grained categories.

As shown in Fig. 2, the main contributions and innovations of this work are summarized as follows.

1) By exploring the main factors limiting the performance of the RSFROD task, we propose aligning and completing detector (ACDet), an efficient single-stage detector. With less computational costs, it surpasses mainstream methods on four public datasets.
2) A plug-and-play A^3M is proposed for locating, arbitrarily oriented, densely arranged rigid objects precisely. It implicitly decouples the centralized bias from the classification branch, thereby aligning anchor points (APs) with ground truths (GTs) adaptively.
3) We propose an OTS algorithm to supplement the incomplete samples in long-tail categories. As a sample assignment postprocess, it can be inserted into existing dense prediction detectors to improve the accuracy of recognizing tail categories.
4) An AGP is designed to enrich feature diversity for recognizing fine-grained objects. It varies feature sampling positions through group learning, achieving richer feature representations with fewer modifications.

II. Related Works

This section mainly discusses the localization alignment and incomplete sample-feature issues in object detection tasks. The latter is subdivided into two aspects: the long-tail problem of samples and the discriminative feature extraction in the RSFROD task.

A. Localization Alignment in Object Detection Task

This section describes localization alignment in object detection tasks, focusing on both natural and remote sensing (RS) scenes.

In the natural scene, classic object detection methods [24], [25], [26] utilize a uniform distribution of anchor boxes corresponding to each bin on the feature map to achieve dense prediction of GTs. This anchor box mechanism uniformly presets several anchor boxes of different sizes and aspect ratios to accommodate all possible situations. The early anchor-free methods [27], [28], [29] simplify the redundant predicted anchor boxes mentioned above. Specifically, it allows each bin on the feature map to predict only one bounding box, significantly reducing computational redundancy. To achieve a more efficient localization alignment process, some recent transformer-based detection methods [30], [31] have successfully eliminated the anchor mechanism and nonmaximum suppression (NMS) processing based on the Hungarian algorithm. However, these methods perform poorly for tiny-sized, densely arranged, and arbitrarily oriented targets.

The bird's-eye-view perspective and the distinctive objects in RS imagery pose significant challenges for precise localization alignment. The methods in [16] and [17] solved the problems of angular alignment between horizontal regions of interest features and arbitrarily oriented targets in traditional detectors. Furthermore, to solve the serious misalignment problem between anchor boxes and axis-aligned convolutional features, the work [32] designed a feature alignment module to alleviate the inconsistency between classification scores and localization accuracy. In addition, the methods in [33] and [34] employed a "from-coarse-to-fine" approach, dynamically improving sample quality by gradually introducing geometric and classification constraints on sample assignment during training. To remove redundant noise and effectively handle

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densely distributed small targets, Zhu et al. [35] proposed a method based on global multilevel perception and dynamic region aggregation. It effectively alleviates the challenges associated with densely distributed targets and enhances the model's discriminative ability at the fine-grained level.

In summary, although many works have made significant progress in localization alignment, they have not directly addressed the contradiction between the uniformly distributed anchors mechanism and the targets that appear in random positions. The exploration and improvement of the anchor mechanism remain a valuable challenge.

B. Long-Tail Samples Distribution in for Task

This section discusses the typical solutions proposed to address the problems caused by the long-tail distribution of samples in natural and RS scenes, respectively.

Zhang et al. [36] proposed a unified distribution alignment strategy for long-tail visual recognition, which provides a flexible and unified solution for various scenarios in visual recognition tasks. Wang et al. [37] designed category-aware angular margin (C2AM) loss, which effectively alleviates the problem of poor performance on tail categories by introducing an adaptive angle margin to redirect the ratio of the classifier weight norm. A new debiasing method, inverse image frequency (IIF) [38], was subsequently proposed, achieving better performance than similar methods in balancing training data. Tan et al. [39] found that the leading cause of the long-tail problem is the imbalanced gradient during model training; therefore, they proposed a gradient-driven training mechanism to alleviate this problem effectively. Liu et al. [40] proposed an active learning scheme based on visual memory, which enables the recognition of long-tail classes in a data-efficient manner.

The fine-grained object recognition task in RS scenarios also exhibits the phenomenon of a long-tail distribution of samples, which affects the model's recognition effect on tail categories. Specifically, Zeng et al. [41] designed a refined instance switching module to alleviate the class imbalance problem. Ouyang et al. [14] built a class-balanced sampler to sample each class with equal frequency. Cheng et al. [20] designed a hierarchical classification paradigm to select and enhance difficult-to-distinguish samples by maintaining multilevel consistency. Zhao et al. [42] proposed a central feature consistency learning mechanism to prevent the model from over-focusing on samples from head categories.

In summary, the current sample imbalance problem caused by the long-tail distribution in natural and RS scenarios is primarily addressed by loss measurement methods that increase the model's attention to tail category samples. Such methods usually require analyzing and processing the entire dataset in advance, which has certain limitations in open data scenarios.

C. Discriminative Feature Extraction in for Task

The insufficient model's ability to extract discriminative features may result in incomplete features and ultimately lead to poor recognition accuracy. Specifically, Chang et al. [43] proposed a mutual channel loss that can guide each feature channel to reflect different local discriminative regions of a specific category. Yi et al. [44] proposed modules that can reinforce each other to purify and enhance basic features, thereby obtaining sufficient discriminative features. Ouyang et al. [14] introduced prototypes to learn the features of fine-grained objects and then utilized contrastive learning to compare the target and learned features, thereby enhancing the distinguishability of fine-grained objects. Li et al. [15] focused on improving the quality of candidate regions and proposed a bilinear channel fusion network to extract independent and discriminative features of proposals. In the fine-grained ship recognition task, Yang et al. [45] combined the attention idea to identify discriminative regions and local features corresponding to subtle visual features. Qi et al. [46] proposed an end-to-end solution for real-time detection tasks for small targets. Specifically, three plug-and-play modules were designed for adaptive up-sampling and multiscale fusion of features separately. To address the issues of limited features, insufficient contextual information, and poor localization ability for small targets, Li et al. [47] designed a model based on the attention mechanism, multiscale feature learning, and small target weighted loss.

In summary, enhancing discriminative feature extraction typically involves increasing the model's learnable parameters and computational complexity. The tradeoff between the two is essential in research.

III. Methodology

A. Problem Setup

First, we feed an image $\mathcal{I} \in \mathbb{R}^{H\times W\times 3}$ into the encoder consisting of backbone and neck structures, as follows:

$$\mathcal{F} = \text{neck}\left(\text{backbone}\left(\mathcal{I}\right)\right) \tag{1}$$

where $\mathcal{F} = \{f_i\}_{i=1}^{L}, f_i \in \mathbb{R}^{H_i\times W_i\times 256}$ represents a list of multiscale feature maps. For instance, we select the standard ResNet series [48] for the backbone and the classic feature pyramid network (FPN) [49] for the neck structure.

Then, we feed $\mathcal{F}$ into a dense prediction head

$$\{\mathcal{F}_{\text{cls}}, \mathcal{F}_{\text{loc}}\} = \text{head}_{\text{den}}\left(\mathcal{F}\right). \tag{2}$$

To obtain the predicted OBBs $\mathcal{P}_{\text{OBB}}$, we first decode the output localization offsets $\mathcal{F}_{\text{loc}}$ with preset anchor boxes $\mathcal{A}_{\text{OBB}}$. This process can be formulated as

$$\mathcal{P}_{\text{OBB}} = \text{decode}\left(\mathcal{F}_{\text{loc}}, \mathcal{A}_{\text{OBB}}\right) \tag{3}$$

where $\mathcal{A}_{\text{OBB}} = \{a_{\text{OBB},i}\}_{i=1}^{L}, (a_{\text{OBB},i} \in \mathbb{R}^{H_i\times W_i\times 5})$ represents the preset uniform anchor boxes, and 5 refers to the dimension of the representation (x_c, y_c, w, h, θ) for an OBB. By the way, the center coordinates contained in the anchor boxes can be represented as APs $\mathcal{A}_{cxy}$.

During the training process, we assign positive and negative samples based on the relationships between the GTs $\mathcal{G}_{\text{OBB}}$ and $\mathcal{P}_{\text{OBB}}$, further optimizing the classification and localization abilities of the model, formulated as

$$\left\{\mathcal{S}_{\text{pos}}, \mathcal{S}_{\text{neg}}\right\} = \text{assign}\left(\mathcal{P}_{\text{OBB}}, \mathcal{G}_{\text{OBB}}\right) \tag{4}$$

where $\mathcal{S}_{\text{pos}}, \mathcal{S}_{\text{neg}}$ represent the sets of assigned positive and negative samples, with each referring to the indices of its corresponding bins on the feature map.

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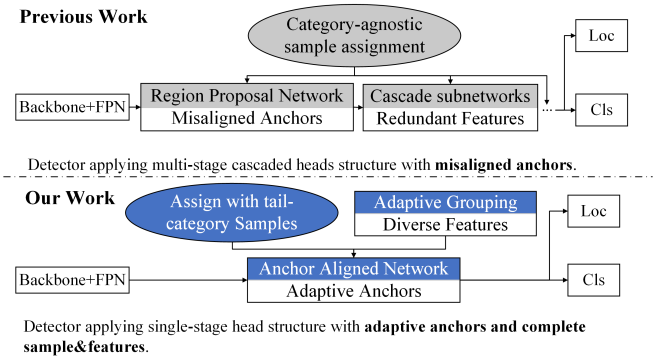


Fig. 2. Comparison between our proposed work and previous work.

Last, the obtained features $\{\mathcal{F}_{\text{cls}}, \mathcal{F}_{\text{loc}}\}$ and positive and negative samples $\{\mathcal{S}_{\text{pos}}, \mathcal{S}_{\text{neg}}\}$ will be applied to calculate the loss function and guide the gradient backpropagation to optimize the model parameters (described in detail in Section III-D).

B. Adaptive Aligning Anchor Mechanism

1) Problem Analysis: The APs-GTs misalignment issue comes from the contradiction between the existing uniform anchor distribution and the random position of targets. It leaves regression branches to learn the offsets between them, increasing the model's learning difficulty. However, in RS scenarios, rigid targets possess arbitrary directions, tiny sizes, and dense arrangements. As a result, they can be arranged in larger numbers within the same area, causing trouble in sample assignment. Specifically, for each positive sample, it becomes increasingly ambiguous to assign a corresponding GT. For each GT, fewer positive samples are assigned. It poses a significant challenge to the learning of offset regression.

To address the above issue, we propose an A^3M, as shown in Fig. 3. It includes an AAN and an AAG. In AAN, the classification and localization branches are highly coupled. It adaptively learns the alignment bias from the categorical features and guides it into the localization branch. AAG collects offsets at the image level and generates adaptive aligned anchors close to their neighbor GT. The above steps do not introduce additional supervision information and can be inserted into the existing dense prediction head structures.

2) Rationale and Advantages: Using categorical features to guide localization bias, we mainly consider two factors: the centripetal sampling of categorical features and the regular structure of rigid objects.

Centripetal Sampling of Categorical Features: In RS images, objects are spatially correlated with their discriminative features from feature maps. Therefore, the categorical features distributed around the target tend to sample toward the spatial position where the target is located, exhibiting centripetal behavior. By leveraging the sampling capability of deformable convolution for feature positions, we successfully decouple the "centralization" process from the classification branch.

Regular Structure of Rigid Objects: In RS scenes, rigid objects usually possess regular geometric shapes. When regular geometric and centripetal forces coexist, the "centralization" process ultimately tends to converge toward the center of the object. Considering that the learning of localization branches involves regressing the anchor box center, length, width, and angle. Therefore, we leverage this trend to offset the misalignment between the preset uniform APs and the targets.

It has several advantages compared to existing solutions as follows.

Compared With Uniformly Distributed Anchors: 1) The matching process from positive samples to GTs is more explicit. 2) The intersection over union (IoU) between positive samples and GTs is effectively improved, reducing the difficulty of model learning. 3) This process does not introduce manually adjusted parameters or additional supervision information.

Compared With Cascaded-Regression-Based Methods: 1) It leverages the hidden optimization goal after coupling with the classification branch, which tends to learn the alignment bias from the object center. 2) It replaces multiple regression processes with a single one, reducing the complexity of regression learning. 3) Its structure is more concise and efficient, eliminating the need for feature alignment and reducing the frequency of encoding and decoding operations on the OBB representation.

3) Mechanism Design: The entire dense prediction head consists of a series of cascaded standard convolutions, AAN, and a classification and localization layer. It can be observed that AAN is inserted between the other two and is composed of several cascaded adaptive aligning modules (AAMs). It is worth mentioning that the traditional dense prediction head is a two-branch, decoupled network where the classification and localization branches are independent of each other.

The module structure of AAM is shown in Fig. 4 (bottom), which takes both the classification and localization branches as input. Specifically, the categorical feature maps are first input into the corresponding deformable convolution. The offsets of the deformable convolution come from two parts. One part is derived from the input features themselves, which adaptively learns the spatial position of the alignment bias through standard convolutions. For the other part, we introduce dilated priors to enlarge the receptive field of the offsets. Next, we input the offsets and localization feature maps into the second deformable convolution to complete the extraction. The input and output of AAM are connected end-to-end and do not share parameters. Finally, we accumulate the offsets output by each AAM and output them together with the classification and localization features output by the last module.

After obtaining the $\mathcal{F}_{\text{off}}$, the next step is to apply the proposed Adaptive Anchor Generation (AAG). Assuming offsets $\mathcal{F}_{\text{off}}$ have the dimension of $\{f^i\}_{i=1}^{L}, (f^i \in \mathcal{R}^{H^i \times W^i \times 2O})$, where O represents the number of offsets. And a list of preset APs $\mathcal{A}_{cxy} = \{a_{cxy,i}\}_{i=1}^{L}, (a_{cxy,i} \in \mathcal{R}^{H^i \times W^i \times 2})$ is also provided. The calculation can be formulated as

$$\mathcal{A}'_{cxy} = \text{sum}\left(\mathcal{A}_{cxy}, \alpha \cdot \text{avg}\left(\mathcal{F}_{\text{off}}\right)\right) \tag{5}$$

where $\text{avg}(\cdot)$ denotes the average operator that is implemented on the last dimension of $\mathcal{F}_{\text{off}}$, and $\text{sum}(\cdot,\cdot)$ denotes the pixel-by-pixel addition, and α is the downsampling rate of the feature map of each layer, calculated as $\{2^{1+i}\}_{i=1}^{L}$. The obtained

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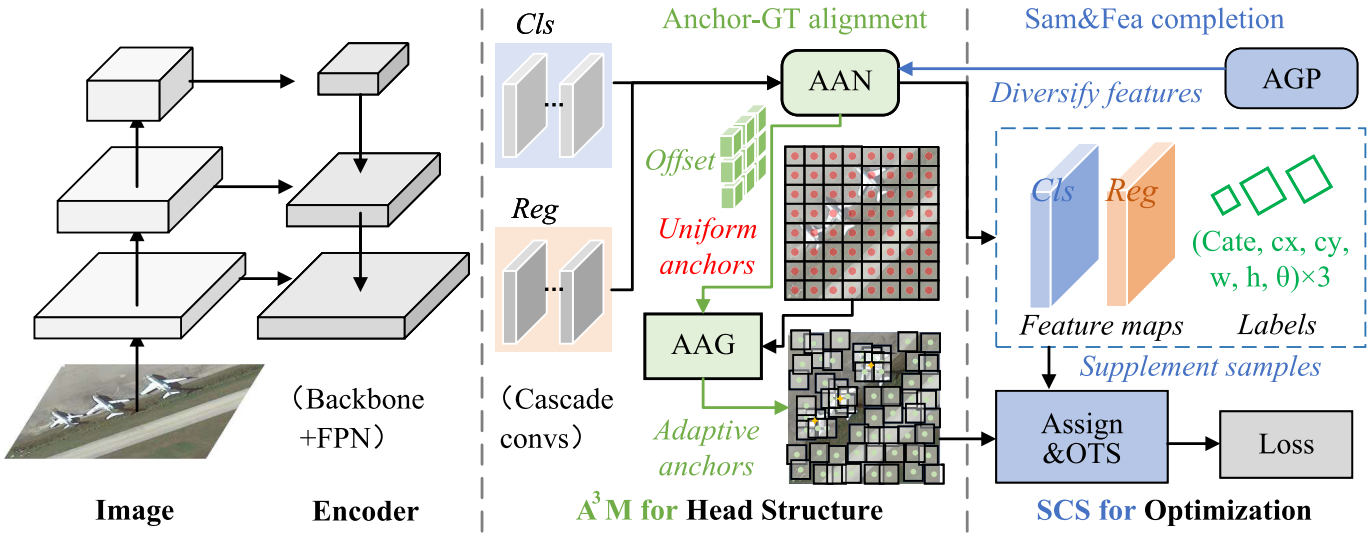


Fig. 3. Architecture of the proposed ACDet during the training process. The image is first input into the encoder to extract multiscale features. In the head structure, the A^{3}M and AGP mechanisms are applied to decode the multiscale features and generate adaptive anchors. In the optimization process, the extracted features, adaptive APs, and GT labels are fed into the sample assignment process with OTS to obtain the final positive samples. Finally, the loss function is calculated based on the positive and negative sample distribution and the GT labels to optimize the ACDet.

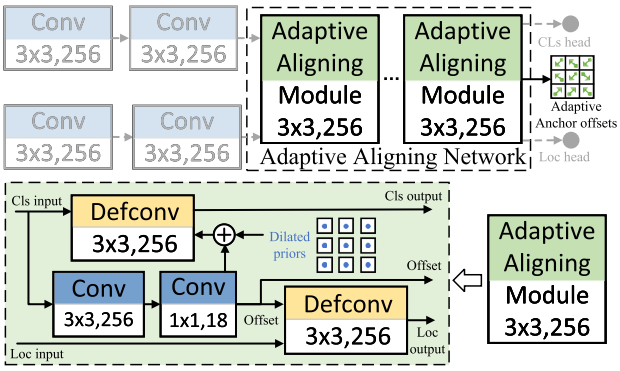


Fig. 4. Structure of the proposed AAN.

adaptive APs $\mathcal{A}'_{cxy}$ will be applied to $\mathcal{A}_{\text{OBB}}$ and be reintroduced into Formula (3) to complete subsequent operations.

A^{3}M ensures that the output anchors move closer to the center of their neighboring targets with an appropriate step. It is worth noting that it introduces acceptable learnable parameters and slightly increases the computational complexity. Next, the generated adaptive APs will serve as the initial positions of the predicted OBBs in the sample assignment and loss calculation process. They will be updated dynamically during the model training process. Sufficient and adequate experiments will demonstrate the effectiveness and simplicity of the proposed A^{3}M (see Section IV-D1).

C. Sample-Feature Completion Strategy

1) Problem Analysis: In RS scenarios, fine-grained categories are usually distinguished based on their usage. For example, the Boeing 747, Airbus A330, and others, in the case of airplanes, as well as motorboats and cargo ships. However, some categories have a low frequency of occurrence due to limited usage scenarios, and the targets of different fine-grained categories often exhibit similar apparent structures. The above situations will lead to two problems: long-tail distribution and slight interclass variance, which affect the model's training process on fine-grained classification tasks. Specifically, the fewer samples in the tail category, the worse the model learning effect is. The model typically has a poor ability to extract discriminative features for targets with a similar appearance. This lack of samples and features is a result of ignoring a key part of the whole rather than simply a lack of quantity. In summary, enhancing the quality and diversity of samples and features is crucial for addressing the issue of sample-feature incompleteness.

Based on the above considerations, we propose a comprehensive sample assignment and feature extraction strategy. First, the single-stage method typically focuses more on defining positive and negative samples but lacks the quality mining of fine-grained category positive samples. We systematically review and analyze the various sample assignment strategies employed by existing detectors and propose a new paradigm. Based on this, we further design an OTS algorithm to enhance the model's focus on tail categories. Second, compared to the fully connected layers used in the multistage detector, the single-stage detector typically employs standard convolutions in the classification and localization branches. As a result, the receptive field is relatively small and fixed, limiting the model's ability to learn and extract discriminative features. We analyze and conclude that it is one of the primary reasons why single-stage methods generally perform worse than multistage methods in the RSFROD task. Based on the above conclusions, we introduce a mechanism named AGP further.

2) Mathematical Modeling: The general paradigm of existing sample assignment strategies can be expressed as follows.

1) *Step 1:* Calculate the relationship between samples and GTs with any metric (IoU between preset anchor and GT [24], IoU between predicted OBB and GT [50],

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Algorithm 1 Algorithm of OTS

Require:
GT OBBs $\mathcal{G}_{\text{OBB}}$ and GT labels $\mathcal{G}_{\text{lab}}$;
positive samples from the previous steps $\mathcal{S}_{\text{pos}}$;
Ensure:
completed positive samples $\mathcal{S}^c_{\text{pos}}$;
initialize $\mathcal{S}^c_{\text{pos}} = \mathcal{S}_{\text{pos}}$;
define $\mathcal{R}_{\text{GT}} = \{\}$;
for each category **do**
  count the GT number based on $\mathcal{G}_{\text{lab}}$, divide it by the total number of GT, derive the ratio r_{GT};
  put r_{GT} into $\mathcal{R}_{\text{GT}}$;
end for
calculate the average number of positive samples per GT $n^{\text{all}}_{\text{ave}}$ based on $\mathcal{S}^c_{\text{pos}}$;
calculate the Intersection over Union IoU between $\mathcal{S}_{\text{pos}}$ and $\mathcal{G}_{\text{OBB}}$;
for each category **do**
  if $r_{\text{GT}} \geq \beta$ **or** $r_{\text{GT}} = 0$ **then**
    break;
  end if
  obtain the average number of positive samples $n^{\text{cat}}_{\text{ave}}$ based on $\mathcal{S}^c_{\text{pos}}$ and the category;
  if $n^{\text{cat}}_{\text{ave}} \geq n_{\text{ave}}$ **then**
    continue;
  end if
  remove the corresponding samples from $\mathcal{S}^c_{\text{pos}}$ based on the category;
  select top $\lceil r_{\text{IoU}} \cdot n^{\text{all}}_{\text{ave}} \rceil$ largest samples based on IoU and the category, append them into $\mathcal{S}^c_{\text{pos}}$;
end for
return $\mathcal{S}^c_{\text{pos}}$;

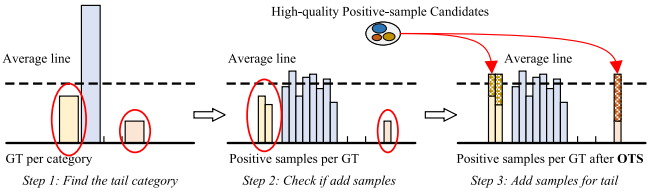


Fig. 5. Brief introduction to the OTS algorithm. It can be roughly divided into three steps. First, it defines the tail categories per image. Second, it checks whether each GT needs to be supplemented. Finally, high-quality sample candidates are supplemented to the corresponding GTs moderately.

generalized Jensen–Shannon divergence [33], predicted classification score [50], and mixture [9], [50], [51]).

2) *Step 2:* Based on the threshold derived from a specific rule (manually set [24] and distributed calculation [52]), obtain positive samples that meet the threshold conditions under the metric.
3) *Step 3:* Filter the remaining positive samples based on the relative position relationship between the sample AP and the GT contour (inside or outside [27] and probability density function [33]).

It can be found that the existing sample assignment strategies for single-stage methods usually lack category constraints. This further affects their performance on RSFROD tasks. In contrast, the multistage model also splits the classification subtask into two steps while cascading regression. They involve binary classification tasks for distinguishing foreground and background, as well as multiclassification tasks for each fine-grained category within the foreground. Therefore, cascade regression can also reduce the learning difficulty of the model for the classification branch, despite requiring a large number of parameters and computational complexity. To compensate for the adverse effects of the problems above, we propose the OTS Algorithm 1. It is designed for the single-stage method and is used to supplement the samples of the tail category online after each round of sample assignment in the training process (see Fig. 5).

In Algorithm 1, $\beta \in [0, 1]$ is used to determine whether a positive sample belongs to the tail category. In Section IV-D2, we will further verify its selection (final $\beta = 0.2$). In step 13

$$r_{\text{IoU}} = e^{\sigma(\beta - r_{\text{GT}})} \tag{6}$$

is a scaling factor of $n^{\text{all}}_{\text{ave}}$ (range in $[1, 1.5]$ approximately) to increase the positive samples number from tail categories. The discussion on its form and σ will be discussed in Section IV-D2. The proposed Algorithm 1 is concise and efficient. In addition, it can be easily integrated into the existing sample assignment strategies to achieve dynamic supplementation of tail samples. Moreover, it is only used during training and does not affect the inference process.

In addition, we introduce the AGP for diversifying features during model training. The design inspiration of AGP comes from the multihead mechanism in self-attention [53]. The core lies in the group learning of offsets and realizing the dynamic expansion of the focusing area under the premise of unchanged parameter quantities. This compensates for the limitation of the single-stage method, which has a poor discriminative feature extraction ability due to its limited receptive field compared to the multistage method. We will next explore how it works and how it can be integrated into existing detection branches. Given the input feature maps $\mathcal{F} = \{f^i\}_{i=1}^{L}, (f^i \in \mathcal{R}^{H^i \times W^i \times 256})$ derived from formula 1, where 256 represents the number of channel dimension, for each layer, we first split it evenly into several groups

$$\begin{aligned} f^i_s &= \text{split}\left(f^i\right) \\ &= \left\{f^i_{s,j}\right\}_j^M \end{aligned} \tag{7}$$

where M should be a positive integer divisible by 256. The choice of M will be further discussed in Section IV-D2. Next, we create M groups of AAMs whose input and output channels of all operators are $256/M$ (except for the 18-D output standard convolution). Please note that each AAM in the group does not share parameters. For each splitted feature map $f^i_{s,j}$, do

$$f^i_{p,j} = \text{AAM}^i\left(f^i_{s,j}\right). \tag{8}$$

Finally, we concatenate M groups of feature maps obtained in each layer in the channel dimension

$$f^i_p = \text{cat}\left(\left\{f^i_{p,j}\right\}_j^M\right). \tag{9}$$

The obtained $\mathcal{F}_p = \{f^i_p\}_i^L$ possess the same dimensions as the input feature maps $\mathcal{F}$. Compared with a single AAM,

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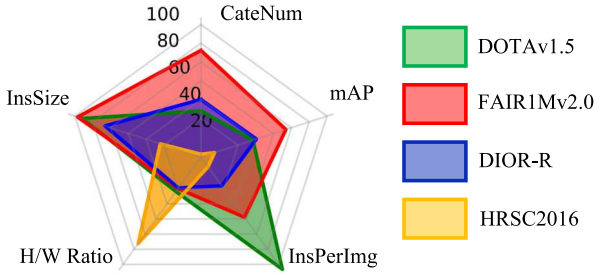


Fig. 6. Radar charts showing the scores of the four datasets on five attributes. Among them, CateNum represents the number of categories, InsSize represents instance size, H/W ratio represents aspect ratio, InsPerImg represents the average number of instances per image, and mAP represents the baseline model performance.

the above process does not introduce additional learnable parameters. The core idea is that group learning can increase the diversity of perspectives without increasing the number of parameters. At the same time, the number of channels of each group from feature maps must be constrained to achieve reliable bias learning. In other words, the choice of M is not the larger, the better. For specific experimental verification, please refer to Section IV-D2.

D. Optimization Process

Based on the obtained features and positive and negative samples, we first calculate the classification loss

$$\mathcal{L}_{\text{cls}} = \text{distance}_{\text{cls}}\left(\{\mathcal{S}_{\text{pos}}, \mathcal{S}_{\text{neg}}\}, \mathcal{F}_{\text{cls}}\right) \tag{10}$$

where $\text{distance}_{\text{cls}}(\cdot)$ select the classic focal loss [24] to optimize the model's classification branch. Next, we introduce the regression learning between the predicted OBBs and GTs based on the positive sample as

$$\mathcal{L}_{\text{reg}} = \text{distance}_{\text{loc}}\left(\text{filter}\left(\mathcal{S}_{\text{pos}}, (\mathcal{P}_{\text{OBB}}, \mathcal{G}_{\text{OBB}})\right)\right) \tag{11}$$

where $\text{filter}(\cdot)$ means filtering out the predicted OBBs corresponding to $\mathcal{S}_{\text{pos}}$ in $\mathcal{P}_{\text{OBB}}$, and $\text{distance}_{\text{loc}}$ selects rotated IoU loss to optimize the localization branch of the model. The last step is to calculate the overall loss

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \gamma \cdot \mathcal{L}_{\text{reg}} \tag{12}$$

where γ is the balancing factor, and the selection will be introduced in Section IV-A2.

IV. Experiments

A. Datasets and Implementation Details

1) Datasets: Experiments are conducted on four classic RS public datasets. To illustrate the characteristics and challenges of the datasets, we analyze their five attributes and visualize them (as shown in Fig. 6). After a simple linear mapping, each attribute value is converted into a score ranging from 0 to 100. It is worth noting that the score is negatively correlated with mAP and instance size, and positively correlated with the others.

1) **FAIR1M-v2.0** [2] provides over 15 000 images with a resolution of 0.3–0.8 m, which comes from various platforms. More than 1 million instances are annotated to five categories with 37 subcategories by OBBs. In Fig. 6, FAIR1M-v2.0 possesses the largest number of fine-grained categories and tiny-sized objects. It is one of the vital bases for verifying the effectiveness of our method.
2) **DOTA-v1.5** [3] includes 2806 images ranging from 800 × 800 to 4000 × 4000 pixels. It covers 188 212 instances with OBB annotations in 15 categories. Based on the comparison in Fig. 6, we observe that DOTA-v1.5 contains relatively densely arranged and tiny objects. Therefore, in the comparative experiments, we will focus on the performance differences between ACDet and other methods in the above cases.
3) **DIOR-R** [4] is a large-scale, public RS dataset with 20 object categories. It contains 23 463 images with 190 288 instances in total, all annotated with OBB format. Compared to DOTA-v1.5, DIOR-R includes more large-scale categories with complex combinatorial structures, such as airports, expressway service areas, and golf courses.
4) **HRSC2016** [54] consists of 1680 images with resolutions cover from 0.4 to 2 m and sizes between 300 × 300 to 1500 × 900 pixels. We uniformly classify all 2976 instances into a single ship category, aiming to observe ACDet's performance in localizing objects with extreme aspect ratios (nearly 5:1).

2) Implementation Details: For model training and testing, a single NVIDIA GeForce RTX 4090 graphics processing unit (GPU) with 24 GB of memory is utilized. Based on MMDetection [55] and MMRotate [56], we can build the model training and testing pipeline. All experiments are conducted by applying the same stochastic gradient descent (SGD) optimizer with a learning rate of 0.0025 on the Pytorch 1.7.0 framework [57]. The learning rate is adjusted using the warm-up and linear decay strategies. For fairness, all datasets are trained on the training set and tested on the validation or test set. All methods employ the same data augmentations, including random horizontal, vertical, and diagonal flips with a probability of 0.25. Note that only HRSC2016 employs a 3× (36 epochs) training schedule, whereas the rest use a 2× (24 epochs) schedule.

Our ACDet utilizes ResNet-50 (loaded with ImageNet [58] pretrained weights) with FPN as the encoder structure, consistent with other compared methods. ATSS [52] is selected for the sample assignment process during model training. Following the same setup as the other methods, we set the balancing factor $\gamma = 1$.

B. Evaluation Metrics

This section presents accuracy and efficiency metrics for evaluating model performance.

1) Accuracy Metrics: We select the mean average precision (mAP) to evaluate the models' performance on the RSFROD task. Precisely, it can be calculated through the following steps.

1) **IoU Calculation:** IoU calculates the ratio of the intersection and union between the predicted OBBs and the GTs as follows:

$$\text{IoU} = \frac{\mathcal{P}_{\text{OBB}} \cap \mathcal{G}_{\text{OBB}}}{\mathcal{P}_{\text{OBB}} \cup \mathcal{G}_{\text{OBB}}}. \tag{13}$$

2) **Sample Statistics:** Based on the results of the IoU calculation, we consider predictions with an IoU of 0.5

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TABLE I

ACCURACY COMPARISONS WITH THE STATE-OF-THE-ART (SOTA) MODELS ON FAIR1M-V2.0 DATASETS

Category	Cascade heads structure (Multi-stage)								Single head structure (One-stage)						Ours
	S2ANet [32]	R3Det [18]	F.-O [25]	G. V. [59]	ORCNN [17]	DRNet [60]	PETDet [15]	ReDet [7]	Retina-O [24]	GWD [61]	KLD [62]	DCFL [33]	FCOS-O [27]	DIMA [20]	
B737	38.9	39.1	36.0	34.2	32.8	40.8	38.1	39.0	36.8	**42.3**	41.0	41.3	42.0	36.7	39.8
B747	79.8	78.7	81.6	77.4	82.0	74.5	83.3	85.4	82.0	81.5	82.6	79.3	77.6	78.3	**85.8**
B777	16.0	17.6	11.2	10.6	13.4	10.8	15.4	17.2	15.8	**22.5**	15.7	14.3	19.6	15.1	15.9
B787	48.6	39.9	41.2	40.6	43.6	45.4	49.0	**57.1**	48.0	49.3	49.0	47.9	49.6	47.2	53.4
C919	0.8	0.2	15.9	15.1	10.9	16.5	13.6	**18.2**	0.7	0.3	0.8	0.4	4.7	0.2	16.4
A220	42.0	42.9	42.2	39.8	39.2	46.9	42.6	42.3	42.7	41.0	40.1	**44.8**	44.1	42.3	42.3
A321	50.5	53.5	50.5	46.2	51.8	51.0	50.0	**57.8**	51.2	54.1	54.7	49.7	55.4	49.1	54.8
A330	45.4	39.8	41.2	35.3	40.1	28.9	47.9	43.2	37.2	41.8	40.5	34.8	42.6	45.9	**45.5**
A350	48.8	49.6	53.2	54.2	59.7	52.0	57.8	44.0	53.2	56.7	62.1	55.4	46.2	54.0	**63.1**
ARJ21	17.1	15.5	10.6	12.6	13.2	19.9	13.3	13.5	16.2	**19.4**	13.1	13.6	15.3	10.2	12.8
PS	19.6	27.7	24.6	25.7	31.2	29.1	31.6	**32.3**	21.5	20.7	22.5	29.7	31.2	31.6	27.4
MB	32.8	41.9	50.5	49.9	**56.9**	55.3	49.5	56.8	32.1	37.8	39.0	37.5	49.6	53.2	53.3
FB	14.9	22.6	20.4	21.0	28.4	28.8	28.3	34.5	11.1	10.6	12.2	22.6	24.5	27.8	**35.5**
TB	21.0	22.8	26.1	**28.8**	27.4	15.5	24.7	19.5	21.7	24.5	24.0	18.9	16.4	26.5	23.1
ES	21.8	32.5	26.7	27.0	29.2	28.7	33.1	45.0	29.8	30.9	32.2	29.9	31.7	32.2	**47.1**
LCS	28.1	34.8	33.4	28.8	**35.0**	36.1	37.4	31.6	22.1	19.7	24.5	32.9	**35.0**	38.3	34.6
DCS	42.3	54.3	50.2	49.9	**60.9**	64.0	59.3	58.7	42.5	42.4	41.5	57.6	56.9	62.2	59.2
WS	27.6	41.1	33.9	34.3	39.2	44.2	47.6	30.7	13.2	14.3	20.9	35.5	32.1	40.8	**43.9**
SC	58.1	**60.8**	56.4	56.4	58.9	58.7	69.5	59.8	49.0	55.4	57.2	57.9	58.4	58.4	59.7
Bus	12.6	14.7	17.4	11.6	**27.2**	28.3	20.6	20.9	13.1	9.4	8.1	15.0	21.5	18.7	19.3
CT	25.8	40.1	43.1	42.0	**44.9**	47.3	46.3	43.6	30.7	31.9	31.3	40.5	42.3	45.0	40.3
DT	31.4	35.4	41.6	41.7	**44.3**	45.0	42.5	44.1	26.8	27.2	25.7	33.8	34.6	42.2	41.0
Van	52.4	56.5	54.0	53.3	56.2	56.5	68.4	57.0	39.4	50.7	53.1	52.6	56.6	55.9	**57.3**
Tri	1.1	10.5	19.7	18.1	**22.0**	16.4	15.2	13.9	2.0	2.4	1.4	12.9	16.6	12.4	13.7
TRC	1.4	0.5	0.6	0.5	1.0	3.4	0.4	1.6	0.4	2.2	1.4	1.6	0.8	0.6	**4.0**
Ex	6.0	16.6	25.7	23.4	**26.8**	33.5	22.7	19.9	12.0	6.9	6.1	23.0	21.1	24.8	21.0
TT	0.4	5.3	22.4	19.1	26.2	36.5	19.6	**28.8**	7.4	2.2	2.0	1.7	13.1	16.4	13.4
BC	43.0	44.8	42.3	42.7	46.5	52.4	45.2	48.4	42.7	43.5	47.0	50.2	47.9	47.1	**50.5**
TC	76.4	80.1	80.7	80.3	81.0	86.9	80.5	86.3	80.6	85.0	80.8	84.4	**85.9**	80.3	80.9
FF	46.9	57.7	56.8	55.7	**62.1**	63.9	62.6	60.4	60.2	61.2	61.2	61.4	60.9	55.1	61.9
BF	84.5	87.4	88.2	80.1	**88.3**	87.4	88.0	87.8	88.1	87.8	87.1	87.2	86.4	84.2	87.4
IS	44.0	48.4	47.8	47.5	48.0	50.7	46.6	45.4	47.1	46.2	45.7	**50.0**	48.3	43.8	49.2
RA	57.9	62.6	63.6	56.4	64.5	55.3	64.0	**72.7**	62.6	60.1	58.8	52.9	59.5	58.7	71.9
BR	15.8	30.4	27.8	23.5	27.3	30.8	27.9	**33.5**	26.8	25.4	26.2	26.0	23.9	23.7	30.3
mAP	33.9	38.4	39.3	37.8	41.8	42.4	42.5	42.7	34.3	35.5	35.6	38.2	39.8	40.0	**42.8**

"B": Boeing, "A": Airbus. "PS": Passenger Ship, "MB": MotorBoat, "FB": Fishing Boat, "TB": Tug Boat, "ES": Engineering Ship, "LCS": Liquid Cargo Ship, "DCS": Dry Cargo Ship, "WS": War Ship. "SC":Small Car, "CT": Cargo Truck, "DT": Dump Truck, "Tri": Trailer, "TRC": Tractor, "Ex": Excavator, "TT": Truck Tractor. "BC": Basketball Court, "TC": Tennis Court, "FF": Football Field. "IS":Intersection, "RA": Roundabout, "BR": Bridge.

or greater as candidates. Next, we count and sort all the obtained candidates based on confidence scores from largest to smallest.

Since the number of GTs in each image varies, we start with the first-ranked candidate and gradually increase the range. Based on the difference between the GTs and the positive samples, we further categorize the candidates into true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs). Precision and recall can be calculated based on the above basis

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{15}$$

3) **Precision and Recall Graphing:** As the range of candidates increases, precision and recall also change accordingly. We record this change in a Cartesian coordinate system with both axes as units. Finally, AP can be derived by calculating the area between the precision–recall (PR) curve and its coordinate axis. If a multicategory task is needed, we define the mean AP of each category as mAP.

2) Efficiency Metrics: For evaluating model efficiency, we selected the number of parameters (Params), GPU memory, frames per second (FPS), and floating-point operations per second (FLOPs) to measure resource usage, inference time, and computational complexity of the model.

1) **Params** indicates the number of parameters (unit: MB) applied during model training (including learnable and nonlearnable parameters). It is used to describe the model size.
2) **GPU memory** counts the graphics card memory (unit: MiB) usage of the model during the training process (batch size is set to 2). Excessive graphics card memory usage will limit the choice of GPU and increase training costs.

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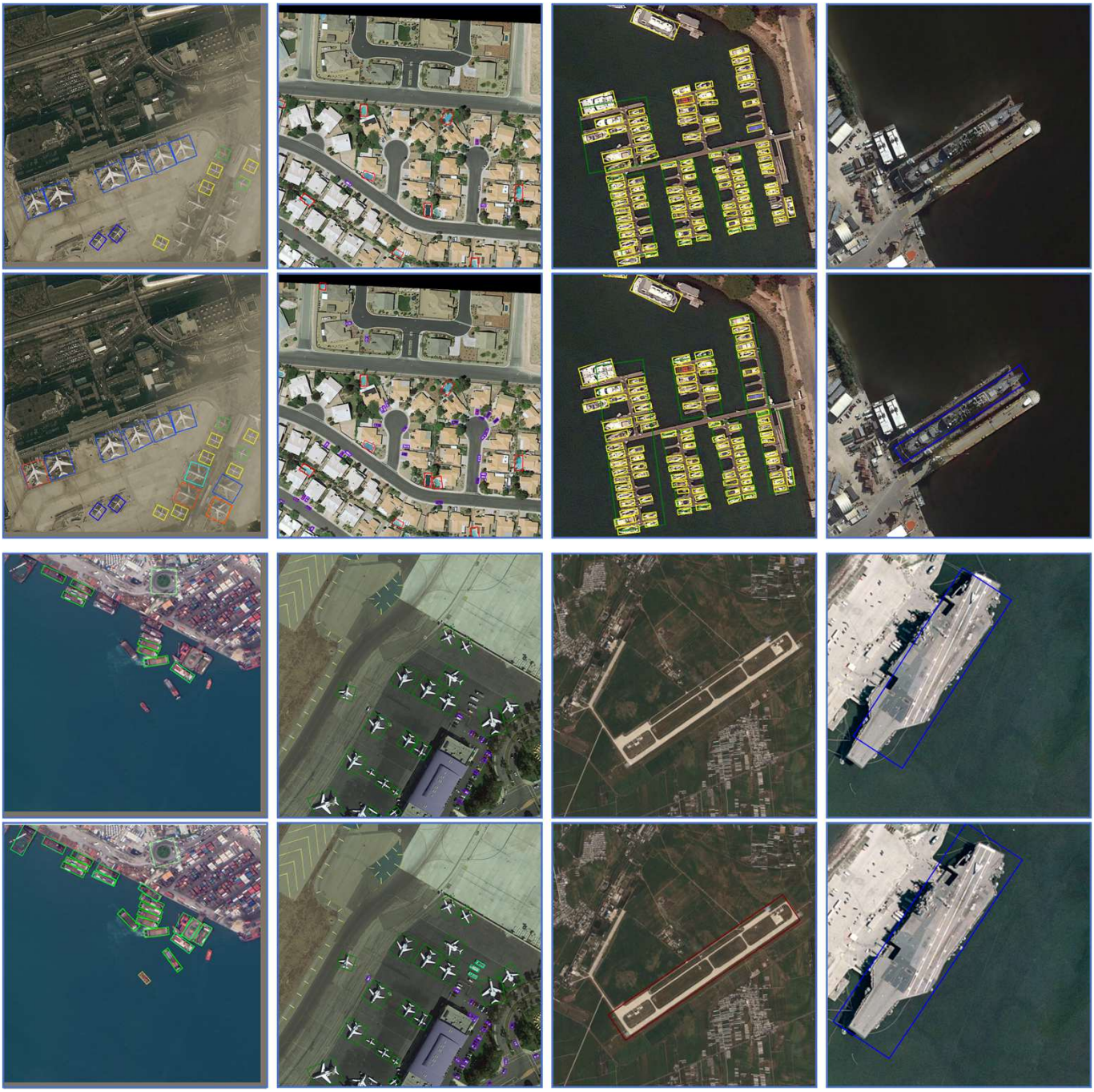


Fig. 7. Visualization of predicted results of the proposed ACDet and baseline (RetinaNet-O). Images from FAIR1Mv2.0, DOTAv1.5, DIOR-R, and HRSC2016 datasets are from left to right. The first and third rows show the results of the baseline model, and the second and fourth rows show those of ACDet.

3) **FPS** represents the number of images the model infers per second. Generally, the larger the number, the higher the inference efficiency of the model.
4) **FLOPs** indicates the number of floating-point operations. We focus on calculating the model's complexity during one forward propagation, with the batch size set to 1.

C. Comparisons With Sate-of-the-Art Methods

1) Accuracy Comparison: We compare the performance of our ACDet with that of the mainstream RSFROD task detectors proposed in recent years on FAIR1M-v2.0, as shown in Table I. They are divided into two camps: multistage methods based on cascaded head structures and single-stage methods based on single-head structures (our method belongs to this category). We also deploy DIMA [20] on the single-stage baseline RetinaNet-O. The difference between the two lies in the number and the connection way of heads used to decode features. Single-stage methods typically feature distinct head structures in both the localization and classification branches. Multistage methods comprise two or more stages in each branch and are connected end-to-end.

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TABLE II
ACCURACY COMPARISONS WITH THE SOTA MODELS ON HRSC2016

Method	RC2 [63]	R^2PN [64]	RRD [65]	RoI T. [16]
mAP	75.7	79.6	84.3	86.2
Method	Gliding V. [59]	R^3Det [18]	S^2A-Net [32]	Ours
mAP	88.2	89.3	**90.2**	89.6

TABLE III
ACCURACY COMPARISONS WITH THE SOTA MODELS ON DOTA-V1.5

Method	**Backbone**	**Plane**	**LV**	**SP**	**mAP**
S^2A-Net [32]	R-50	52.0	69.3	70.4	60.3
Retina-O [24]	R-50	69.0	78.9	63.6	61.7
FPNFormer [66]	R-50	74.0	80.6	64.2	64.2
DCFL [33]	R-50	75.3	80.5	67.5	66.2
FRCNN-O [25]	R-50	71.4	80.2	61.9	64.6
RoI T. [16]	R-50	75.8	81.1	62.7	65.9
PETDet [15]	R-50	**77.3**	**89.9**	70.8	**68.2**
Ours	R-50	77.0	88.9	**71.5**	67.8

For DOTA-v1.5, "LV": Large Vehicle, and "SP": Swimming Pool.

TABLE IV
ACCURACY COMPARISONS WITH THE SOTA MODELS ON DIOR-R

Method	**Backbone**	**Airport**	**Dam**	**TS**	**mAP**
Ret.Net-O [24]	R-50	25.5	20.6	47.9	58.5
FPNFormer [66]	R-50	37.9	25.9	53.8	63.0
DCFL [33]	R-50	51.0	35.7	57.8	66.0
FRCNN-O [25]	R-50	29.1	25.1	54.7	61.8
PETDet [15]	R-50	48.5	34.0	55.0	65.1
Ori. Rep. [67]	R-50	50.3	33.8	56.2	65.4
Ours	R-50	**55.3**	**37.4**	**60.1**	**67.8**

For DIOR-R, "TS": Train Station.

TABLE V
EFFICIENCY COMPARISONS WITH THE CLASSIC MODELS ON DOTA-V1.5. THE FIRST GROUP (TWO ROWS) CONTAINS CLASSIC METHODS, BELONGING TO SINGLE-STAGE AND MULTISTAGE DETECTORS, RESPECTIVELY. THE SECOND ARE COMPETITIVE METHODS, BOTH OF WHICH BELONG TO MULTISTAGE DETECTORS

Method	**Params**	**GPU memory**	**FPS**	**FLOPs**
Ret.Net-O [24]	36.87	8766	21.36	225.88
FRCNN-O [25]	41.16	15664	23.91	225.26
ReDet [7]	**31.80**	19702	16.68	**108.44**
PETDet [15]	46.79	16844	25.38	157.75
Ours	38.04	**7184**	**26.51**	161.06

As shown in Table I, our ACDet achieves the best performance with an mAP of 42.8. Specifically, it exceeds other same-type single-stage methods, surpassing the baseline RetinaNet-O by nearly 8.5 mAP and the second DIMA by about 2.8 mAP. Moreover, it gains a slight advantage over the three latest competitive multistage detectors, ReDet, PETDet, and DRNet. Combined with the visualization in Fig. 7, it reveals that ACDet is more effective in fine-grained object recognition and densely arranged object localization relative to the baseline model. The above results effectively demonstrate ACDet's superior detection capabilities for fine-grained, small-scale, densely arranged objects in RS scenes.

In addition, we conduct comparative experiments on three other datasets: DOTA-v1.5, DIOR-R, and HRSC2016. On DOTA-v1.5, we focus on comparing the differences between ACDet and other methods in detecting tiny and densely arranged objects (planes, large vehicles, and swimming pools). On DIOR-R, we list relatively large-scale categories, including airports, dams, and train stations. Finally, we observe the proposed method's localization ability for targets with extreme scales in HRSC2016. The experimental results verify our ACDet's ability to detect and classify various extreme targets in complex RS scenes.

As shown in Tables II–IV, our ACDet achieves the best performance on DOTA-v1.5 (about 6.1 mAP higher than the baseline model) and DIOR-R (about 9.3 mAP higher than the baseline model), and is second only to S^2A-Net on HRSC2016 (with a difference of 0.6 mAP). Specifically, it outperforms other methods in detecting targets of various scales, including tiny, densely arranged, large, and extreme targets. The second to fourth columns in Fig. 7 also provide a visual comparison. It can be demonstrated that our proposed sample alignment and completion strategies yield stable and robust performance gains for detecting various targets in RS scenes.

2) Efficiency Comparison: We also compare the proposed ACDet with other methods in terms of their training and inference efficiencies. First, we list two classic methods, RetinaNet-O (single-stage) and Faster R-CNN-O (multistage), as baselines to provide standard References. Second, we select two models with suboptimal performance, PETDet and ReDet, to compare their efficiency with that of the baseline and ACDet, respectively. Specifically, GPU memory and parameters are used to assess the computational resource consumption of the model during and after training, respectively. FPS and FLOPs are used to evaluate the model's processing efficiency during the inference process.

Based on Table V, we observe the following phenomena.

1) **Params:** Compared with multistage methods, single-stage methods do not require RPN, RoI align, or fully connected layers (which usually occupy many parameters). Therefore, multistage methods (Faster-RCNN-O and PETDet) usually require more parameters. Excluding ReDet, this is because it employs a lightweight encoder architecture that differs from other models. Our ACDet possesses lower Params than other multistage methods due to its fully convolutional network structure.
2) **GPU Memory:** It is almost half of the multistage methods in single-stage methods. Similarly, due to a more streamlined network structure and a single sample assignment process. The core module in ReDet, due to

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TABLE VI

ABLATION STUDIES OF THE PROPOSED ADAPTIVE ALIGNING ANCHOR MECHANISM. IN THE MAP COLUMN, THE LEFT AND RIGHT SIDES REPRESENT THE PERFORMANCE ON FAIR1M-V2.0 AND DOTA-V1.5, RESPECTIVELY

Method	Dil.Rate	Conv.Num.	Mod.Num.	mAP	Params
baseline	-	-	-	34.3|61.7	36.9
	1			+1.9|+2.3	+1.1
	3			+2.4|+2.6	+1.1
		0		+0.2|-0.8	0
		2		+3.0|+3.2	+2.3
			1	+1.3|+2.1	-0.6
			3	+3.2|+3.6	+2.9
A^3M	2	1	2	+2.6|+3.0	+1.1

"Dil.Rate": priors Dilation Rate in AAN, and "Conv.Num.": standard Convolutions Number in AAN, and "Mod.Num.": Modules number of AAN.

its "rotational equivariance" characteristic, occupies a significant amount of GPU memory. ACDet possesses the lowest memory usage, reflecting its efficient internal operating mechanism.

3) **FPS:** Both single-stage and multistage methods can achieve relatively high levels. ReDet results in low FPS due to its frequent memory access and low parallel computation. ACDet is the fastest, indicating a reasonable collaborative efficiency among its internal modules.
4) **FLOPs:** The large number of anchors preset by RetinaNet-O (typically nine times more than other methods) results in a significant increase in FLOPs. ReDet has the lowest FLOPs, mainly due to its lightweight encoder architecture. But its module has low compatibility with the underlying hardware, so it is not contradictory that it has both low FPS and FLOPs. While ACDet demonstrates reasonable FLOPs as a single-stage method.

D. Ablation Studies

In this section, we examine the contributions of each module in the proposed method. The specific ablation experiments are divided into two: one for A^3M and the other for SCS.

1) Ablation of A^3M: Parameter tuning: For A^3M, we mainly perform ablations in three aspects, as shown in Table VI. The baseline model is RetinaNet-O, and the datasets are FAIR1Mv-2.0 and DOTA-v1.5. Specifically, in a single AAM, we verified experiments with different dilation rates and the number of 3×3 cascaded standard convolutions. In addition, we verified the impact of the number of cascaded AAMs on performance. In each row, the blank area indicates the default configuration, as specified in the bottom row. For the evaluation metrics, we select the mAP and Params to verify the accuracy and efficiency of the proposed module, respectively.

Specifically, the following results can be observed from Table VI, Figs. 8 and 9. 1) After deploying the proposed A^3M, we achieve further performance improvement and faster convergence speed compared to the baseline model with fewer parameters. 2) Increasing the dilation rate within a specific range, the performance on FAIR1M-v2.0 continues to improve, but on DOTA-v1.5, it first increased and then decreased. 3) The number of standard 3×3 convolutions in AAM has a significant improvement on the model, but it also increases the number of parameters. 4) Increasing the number of AAMs can also improve the model performance, but it requires more parameters as well.

The above results provide a quantitative analysis of the impact of selecting each hyperparameter in A^3M on model performance. In addition, the visualization results in Fig. 8 and the loss convergence curves in Fig. 9 further demonstrate its effect on anchor localization, alignment, and model training efficiency.

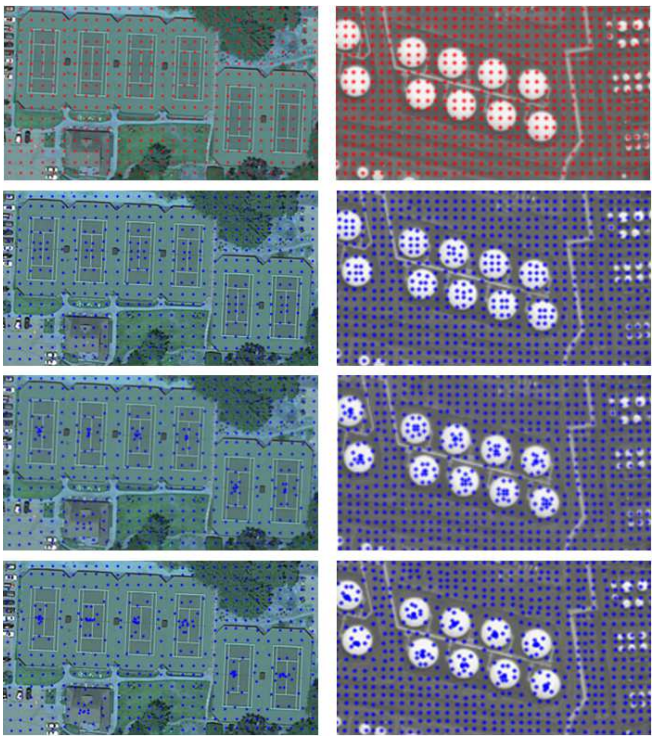


Fig. 8. Visual comparison of anchors. The first row shows the traditional uniformly distributed anchors. The second, third, and fourth rows show the adaptive anchors generated after applying A^3M with AAM numbers 1, 2, and 3, respectively. It can be observed that the adaptive anchors can converge to the center of the target, while the background area is relatively irregular. This convergence effect becomes more obvious as the number of AAMs increases.

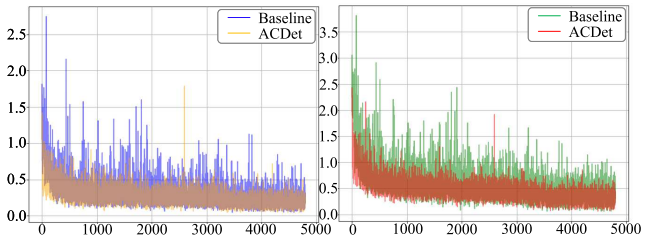


Fig. 9. Comparison of the optimization loss curves of ACDet and baseline (RetinaNet-O). The left figure shows the regression loss (bounding box localization), and the right figure shows the overall loss. It can be observed that, compared to the baseline model, ACDet has a faster convergence speed in both the regression loss and the overall loss.

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TABLE VII

PERFORMANCE IMPROVEMENT OF A^{3}M APPLIED ON DIFFERENT DETECTORS. THE EXPERIMENTS ARE CONDUCTED ON FAIR1M-V2.0, DOTA-V1.5, DIOR-R, AND HRSC2016 DATASETS

Method	A^{3}M	FAIR1M	DOTA	DIOR-R	HRSC2016
Ret. Net-O	×	34.3	61.7	58.5	62.8
	✓	+2.7	+3.6	+3.1	+10.8
ATSS-O	×	38.9	63.0	61.4	78.1
	✓	+1.6	+3.3	+4.4	+10.3
FRCNN-O	×	39.3	64.6	61.8	82.9
	✓	+0.8	+2.2	+3.9	+6.4

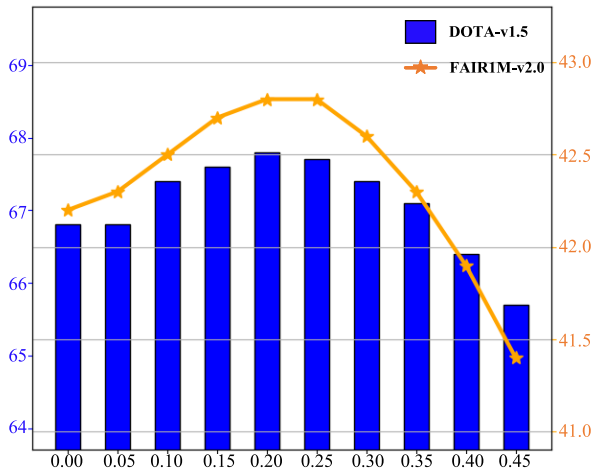


Fig. 10. Ablation study of the hyperparameter β. The horizontal axis represents the value of β, and the vertical axis represents mAP.

Module comparison: We apply the proposed A^{3}M to three classic detectors on the DOTA-v1.5 dataset to verify the effectiveness and generalization of the proposed mechanism, as shown in Table VII. It can be observed that the performance of the two single-stage methods, RetinaNet-O and ATSS-O, has improved by more than three mAP. Specifically, there are significant improvements in the three categories of small vehicles, ships, and helicopters, which are of relatively small size and densely arranged. In particular, we integrate A^{3}M into the multistage detector Faster-RCNN-O and observe an improvement of approximately 2 mAP. Specifically, we introduce A^{3}M into RPN to provide candidate regions with higher localization quality for subsequent stages. This demonstrates the effectiveness and generalization ability of the proposed A^{3}M in the existing mainstream detectors on the localization subtask.

2) Ablation of SCS: Parameter tuning: For OTS, we conduct ablation experiments on two hyperparameters, β and σ. Specifically, β controls whether a GT belongs to the tail category, while σ determines the number of supplemented samples for the tail categories. Fig. 10 illustrates the effect of β on model performance as it varies in the range $[0, 0.5]$ with a step size of 0.05. It can be seen that both datasets show a trend of increasing first and then decreasing. This is because enlarging the tail-category samples range with a suitable proportion is conducive to alleviating the learning imbalance problem caused by the long-tail effect. However, as the selected range expands, the introduction of some nontail-category samples may cause the opposite effect. Finally, considering the comprehensive performance of the two datasets, we selected $\beta = 0.2$.

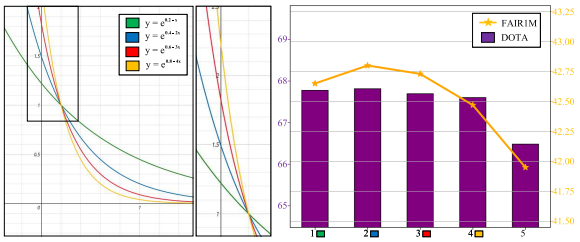


Fig. 11. Under the condition of $\beta = 2$, σ curves and experimental results. The left figure represents the magnification of the supplementary sample at $\sigma = 1, 2, 3, 4$. The right one represents the variation curve of accuracy on two datasets under the corresponding values of σ.

TABLE VIII

ABLATIONS OF THE PROPOSED AGP. WE MAINLY ABLATE THE HYPERPARAMETER M (NUMBER OF GROUPS) AND THE NUMBER OF APG LAYERS. MAP, FLOPS, AND FPS ARE ADOPTED FOR PERFORMANCE TRADEOFF METRICS

Method	M	Layer Num.	mAP	FLOPs	FPS
baseline	1	0	39.5|66.3	159.51	28.01
	2	1	+0.3|+0.4	+0.45	-0.58
	8	1	+0.2|-0.1	+4.27	-4.24
	16	1	-0.3|+0.4	+9.70	-7.66
	32	1	+0.5|+0.4	+20.57	-16.27
	4	2	+0.4|+0.2	+3.85	-3.81
AGP	4	1	+0.6|+0.4	+1.55	-1.50

"M": groups number, and "Layer Num.": layer number.

Next, we optimize σ to provide moderate positive samples for the GTs of the tail categories. In theory, the amount of supplemented positive samples is related to the rarity of tail category samples. That is, the rarer the tail category samples, the more positive samples should be added to gain sufficient attention during training. Therefore, we design a dynamic IoU threshold to supplement positive samples for the tail categories reasonably (as mentioned in Algorithm 1, line 18)

$$r_{\mathrm{IoU}} = e^{\sigma \cdot (\beta - r_{\mathrm{GT}})} \tag{16}$$

where r_{IoU} represents the scaling factor to control the number of supplemented samples. Their relationship is plotted in Fig. 11, left, where σ is a factor that scales the number of supplementary samples exponentially. To explore in more detail the impact of σ selection on model performance, we conduct the following experiments.

It can be observed that as σ increases, both datasets exhibit convex changes (see Fig. 11, right). It demonstrates that increasing the number of supplementary samples moderately is beneficial, while excessive supplementation is harmful. In addition, the overall variation intervals on FAIR1M and DOTA are both approximately 1 mAP, demonstrating the robustness of σ selection.

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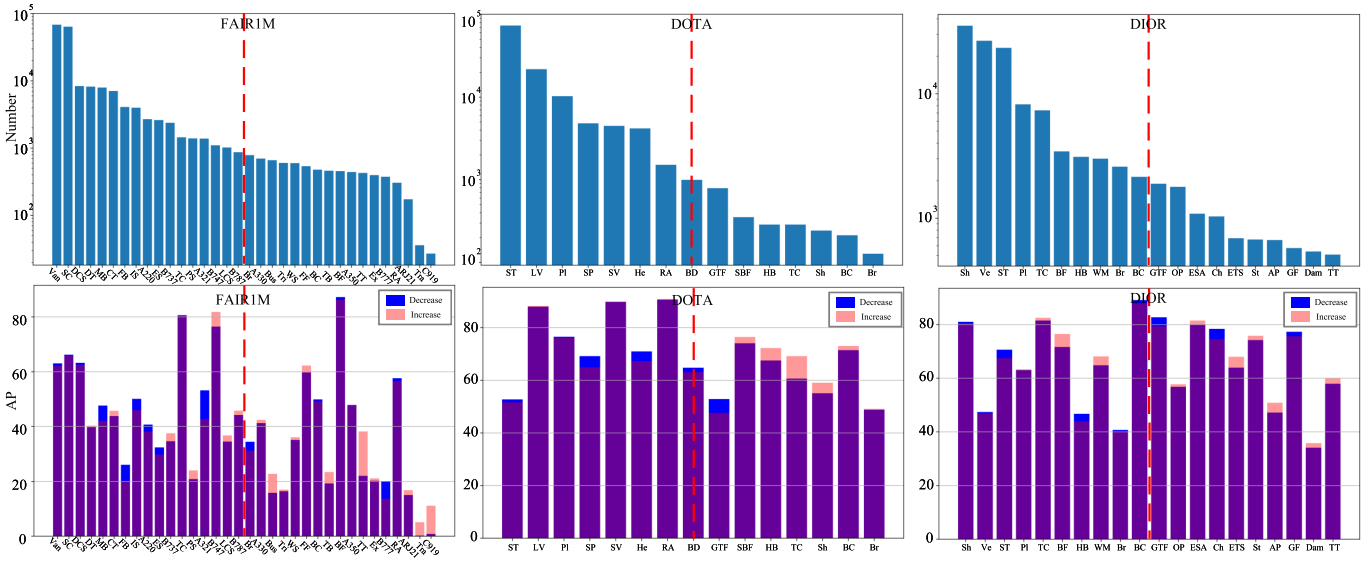


Fig. 12. Impact of OTS on the performance of long-tail categories. The first row presents histograms of the number of categories in the three datasets. The second visualizes the impact of adding OTS to the baseline on the performance of each category.

TABLE IX
PERFORMANCE IMPROVEMENT OF OTS APPLIED AFTER VARIOUS SAMPLE ASSIGNMENT STRATEGIES. THE EXPERIMENTS ARE CONDUCTED ON FAIR1M-V2.0, DOTA-V1.5, DIOR-R, AND HRSC2016 DATASETS

Assigner	OTS	FAIR1M	DOTA	DIOR-R	HRSC2016
MaxIoU	×	34.3	61.7	58.5	62.8
	✓	+1.1	+0.6	+0.5	+0.1
ATSS	×	38.9	63.0	61.4	78.1
	✓	+1.3	+0.5	+0.3	-0.2
C2F	×	37.1	62.2	62.0	80.5
	✓	+1.0	+0.4	+0.6	-0.1

For AGP, based on Table VIII, we can draw the following conclusions. 1) As the number of heads increases, FLOPs and FPS increase exponentially. 2) Similarly, mAP increases appropriately, with fluctuations. We analyze and conclude that since the number of feature map channels is constant (C = 256), an increase of M results in a reduction in the number of channels in each head. Furthermore, the reduction of channel numbers will lead to an incomplete feature extraction problem. In addition, the computational complexity and the number of heads exhibit an exponential growth trend. Therefore, considering the dynamic balance of the above two factors, we ultimately choose the last row (M = 4, layer number = 1) setting.

Module comparison: After completing the selection of β and σ, we connect OTS under various sample assignment strategies. As shown in Table IX, after adding OTS to the three sample assignment strategies, there is a noticeable improvement in averaging all the categories. ACDet shows steady improvements on three public multicategory datasets except for HRSC2016. Considering that the model only categorizes ships on the HRSC2016 dataset, OTS is not triggered by default. The above phenomenon verifies the effectiveness and generalization ability of the proposed OTS.

To further explore the performance improvement of OTS on various categories of datasets, especially long-tail categories, we have conducted several experiments. Specifically, we visualize the sample number and model performance on three datasets for each category, as Fig. 12 shows. We use RetinaNet-O as the baseline and insert OTS after its default sample assigner.

The conclusions are summarized as follows.

1) We observe similar long-tail distributions on all three datasets (FAIR1M, DOTA, and DIOR).
2) The introduction of OTS typically improves the model's performance for the bottom 50% categories. It is attributed to its online supplementation mechanism for tail-category samples, which helps the model focus more on tail categories.
3) The performance of the top 50% of categories has decreased. Since OTS reduces the model's attention to existing positive samples, it also reduces the model's ability to distinguish head categories.
4) It is worth mentioning that OTS improves the overall performance of the baseline by approximately 0.5 mAP (across all categories and datasets).

The above conclusion verifies the effectiveness of OTS as a postprocessing method in improving the performance of the tail categories.

E. Robustness Analysis

1) Localization Stability of A^3M: **First**, we provide a comparison of the performance of ORCNN with and without A^3M at IoU = $\{0.5, 0.6, 0.7, 0.8, 0.9\}$. Specifically, it is inserted into the RPN, providing adaptive APs to replace the uniform APs. As the IoU threshold gradually increases, the judgment for localizing GTs becomes increasingly stringent, directly testing whether A^3M has introduced severe oscillations.

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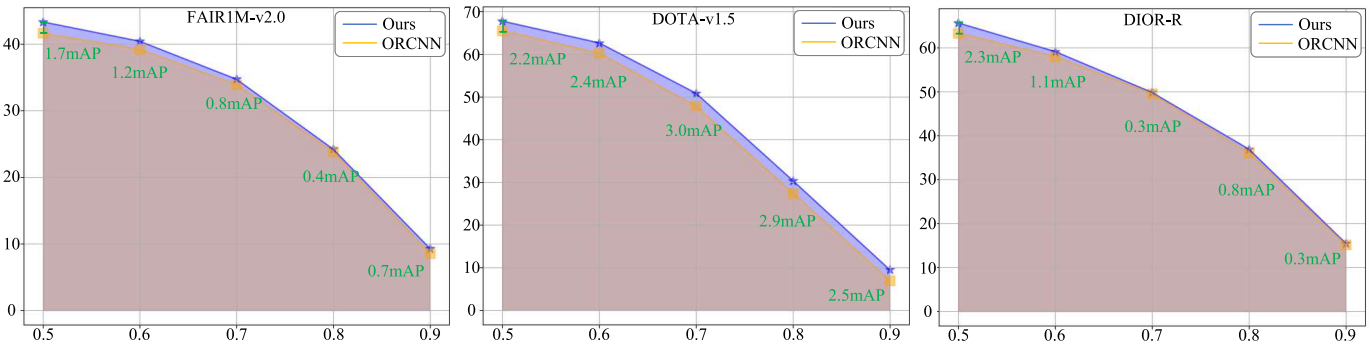


Fig. 13. Performance change curves of two models under different IoU thresholds. The horizontal axis represents the IoU threshold, and the vertical axis represents mAP.

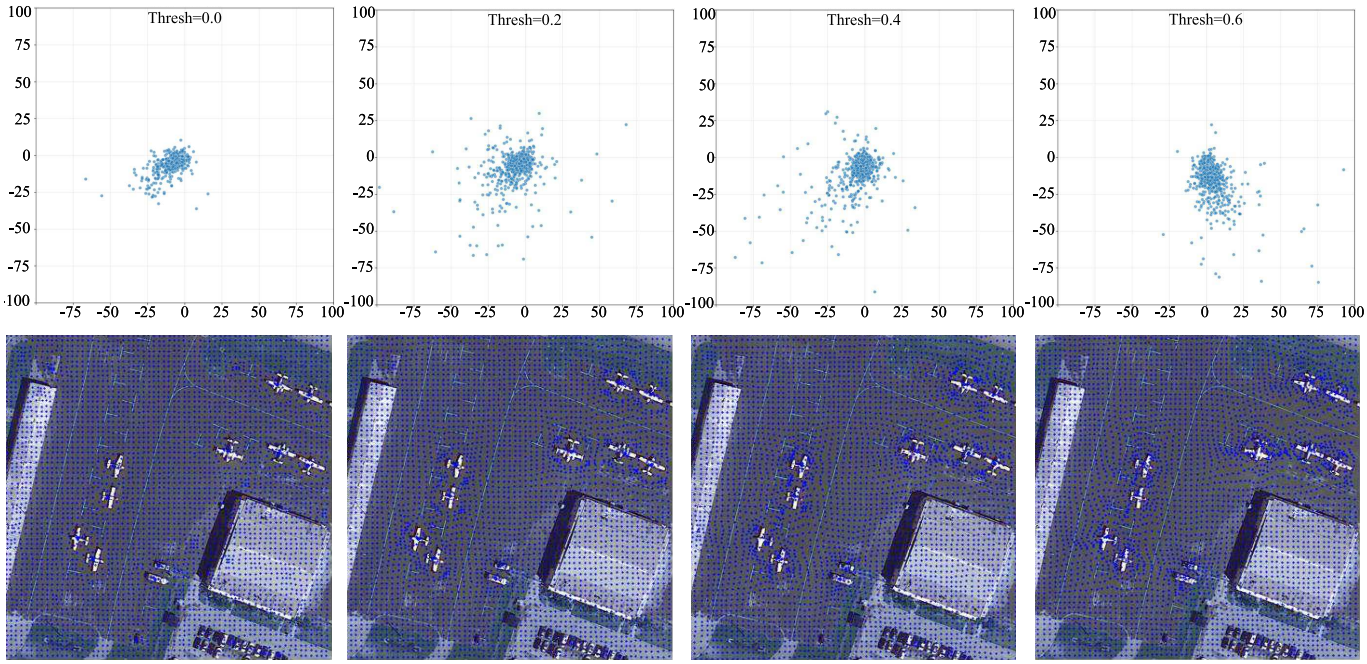


Fig. 14. Visualization of positive sample anchor offset under different label perturbation rates. From left to right, the perturbation rates are 0%, 20%, 40%, and 60%, respectively. The first row displays the overall distribution of adaptive anchor offsets from the DOTA-v1.5 dataset, plotted on the X- and Y-axes. The second row visualizes a single image and its adaptive APs distribution.

Fig. 13 reports the performance differences. We observe that A^{3}M enhances the model's performance in all cases. Although the improvement effect gradually weakens as the IoU threshold increases. A possible reason is that stricter judgment limits significant performance improvement. In addition, we find stable and noticeable improvements on the DOTA dataset (with an average advantage of 2.6 mAP). The above results indicate that as the IoU threshold increases, the introduction of A^{3}M further improves the model's performance, demonstrating its stability powerfully.

Next, we explore whether the offsets may converge to a stable mode.

We introduce random perturbations of localizing labels to observe whether they affect the model's ability to learn and align offsets. Specifically, we randomly translate the GT labels on both horizontal and vertical axes with a certain probability, and randomly sample the offset within the object scale range. Theoretically, incorrect labels may guide the model to learn nonfixed localization patterns, thereby affecting the learning effectiveness of offset alignment. The selection of probability varies in [0%, 20%, 40%, 60%].

It can be observed that as the probability of disturbance gradually increases, the range of adaptive anchor offsets slightly increases, but remains in a nearly circular distribution at the center. On the second row of Fig. 14, we also observe some interesting phenomena.

1) Under undisturbed conditions, adaptive APs within the label range converge toward the center of the target.
2) As the disturbance increases, the number of offset APs increases and spreads around the target.
3) These offset APs exhibit a pattern in which ***the target-inside points move toward the center, while the outside points move away from the target.***

Based on the design concept of the decoder head in A^{3}M, this aligns with the model's learning mode for target categories. Sampling is used to gather central features classifying objects, rather than irrelevant background features.

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TABLE X
IMPACT OF VARIOUS DATA AUGMENTATION STRATEGIES ON ACDET ON DOTA-V1.5. *REPRESENTS ROTATION DITHERING, AND † REPRESENTS MULTISCALE CROPPING

Models	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
ORCNN	76.7	63.4	55.0	49.0	89.8	81.3	62.6	70.3	73.1	50.0	**77.4**	90.7	73.6	62.4	74.2	65.6
ORCNN*	77.1	64.9	54.2	52.9	90.1	88.4	**68.1**	**72.4**	78.0	50.5	76.7	**90.8**	**74.2**	69.8	76.7	68.4
ORCNN*†	**80.8**	**69.8**	**67.2**	**55.8**	**90.2**	**90.1**	54.1	71.4	**80.9**	**61.9**	76.6	**90.8**	73.3	**77.5**	**78.4**	**70.0**
ACDet	77.4	64.7	58.4	49.1	89.9	89.8	53.6	74.6	78.2	54.9	77.8	90.8	**74.7**	70.8	77.3	67.8
ACDet*	78.5	67.3	55.4	53.2	90.0	90.0	54.6	**79.0**	81.1	56.8	**82.4**	**90.9**	74.6	74.1	77.5	69.4
ACDet*†	**81.3**	**68.6**	**68.5**	**54.5**	**90.4**	**90.2**	**70.3**	76.4	**82.6**	**57.3**	79.5	90.8	74.3	**77.9**	**78.3**	**71.4**

"PL": Plane, "BD": Baseball Diamond, "BR": Bridge, "GTF": Ground Track Field, "SV": Small Vehicle, "LV": Large Vehicle, "SH": Ship, "TC": Tennis Court, "BC": Basketball Court, "ST": Storage Tank, "SBF": Soccer Ball Field, "RA": Roundabout, "HA": Habor, "SP": Swimming Pool, "HC": Helicopter.

2) Discussion on the Applicability of A^{3}M/SCS: In this section, we mainly discuss the applicability of RSFROD task methods under two mainstream paradigms on A^{3}M and SCS, and provide theoretical compatibility guarantees.

The DETR families [21], [22] are based on the "set prediction" paradigm, treating object detection as a direct sparse set prediction problem. There are no preset, densely arranged APs, so it does not belong to the "dense prediction" paradigm. Therefore, A^{3}M and SCS cannot be adapted. However, in RSFROD tasks, the performance of the DETR family is typically weaker than that of dense prediction methods due to a mismatch between their design intent and the characteristics of the objects. For DETR families, predicting densely arranged, arbitrarily oriented objects in RSFROD tasks remains a significant challenge.

The anchor-free methods mainly include anchor-free box methods represented by the FCOS [27] and keypoint extraction methods represented by the O2D-Net [23]. Although anchor-free methods, such as FCOS, eliminate the preset anchor boxes, they still rely on APs to initialize samples aligned with the multiscale feature maps. Therefore, it is essentially a dense prediction method, so it also supports A^{3}M/SCS. The keypoint extraction method is not mainstream in RSFROD tasks due to the fuzzy localization caused by keypoint adhesion.

In summary, A^{3}M/SCS currently supports detection methods based on the "dense prediction" paradigm, which are commonly present in RSFROD tasks. In the future, further exploration is needed to adapt A^{3}M/SCS based on the "set prediction" paradigm (DETR series) and keypoint extraction methods.

3) Impact of Data Augmentations on ACDet: As shown in Table X, we conduct experimental verification of the ACDet and ORCNN methods on the DOTA-v1.5 dataset using the three strategies. Among them, the rotation dithering strategy comes from the mmrotate framework. The multiscale strategy involves cropping the training and testing data in a ratio of $[0.5, 1.0, 1.5]$. We include ORCNN to observe and compare the performance gains of ACDet against other methods.

The experimental results demonstrate the performance improvements of ACDet and ORCNN in various categories for comparison. It can be observed that both rotation dithering and multiscale cropping can effectively improve the performance of the two methods. Specifically, the former increases by an average of 2.2 mAP, while the latter increases by 1.8 mAP. It fully demonstrates that effective data augmentation strategies can further improve the performance of ACDet. Based on Table X, we have analyzed and drawn the following conclusions:

1) Rotational jitter can increase the angle range of the targets and improve the ACDet's learning generalization of OBBs (such as ground track field, tennis court, basketball court, and swimming pool).
2) Multiscale cropping can enrich the scale of targets, thereby improving the detection performance of models for giant and tiny scaled targets (such as baseball diamond, bridge, ship, and swimming pool).

V. CONCLUSION

This article proposed an efficient single-stage detector, ACDet, with two plug-and-play modules. It is dedicated to accurately locating and correctly classifying fine-grained rigid objects that are densely arranged and arbitrarily oriented. Specifically, the proposed A^{3}M improves the object's localization learning efficiency in RS scenarios. It alleviates the severe APs-GTs misalignment problem caused by the characteristics of the targets, significantly improving model performance by elegantly decoupling categorical feature bias. In addition, the proposed SCS enhances the model's ability to recognize tail categories and extract discriminative features in fine-grained targets. It gains improvements with low computational resource consumption without modifying the existing structure. Four datasets, FAIR1M, DOTA, DIOR-R, and HRSC2016, were selected for the experiments to verify the proposed method's performance in various scenarios. Experimental results demonstrated that ACDet achieves the SOTA level on three datasets, ranking second on HRSC2016, with the characteristics of the objects mentioned above. Experiments also indicated that the two proposed modules possess strong robustness and generalization. It is worth noting that it achieves faster inference speed and comparable performance to multistage methods, with fewer parameters, lower computational complexity, and reduced training costs.

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